

Aspire Project Report

A Study on Projective Modules

submitted by

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Certificate

This is to certify that the project titled “**A Study on Projective Modules**” that is being submitted by Mr. Harikrishnan T R to Indian Institute of Space Science and Technology, Thiruvananthapuram, is a bonafide record of study carried out by him under my supervision and guidance at the Department of Mathematics, during 12th June 2023 - 12th July 2023.

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Introduction

Projective modules are the direct summands of a free module. They generalize the concept of free modules. Projective modules are important algebraic objects in commutative algebra and K - theory. This project attempts to examine the structure theory of finitely generated projective modules. This project also focuses on the study of an important class of projective modules called stably free modules. We begin with the characterizations and basic results concerning projective modules and proceed to a detailed study on projective modules of rank 1.

The first section, focus on the characterizations and fundamental results related to projective modules. The second section is dedicated to proving the freeness of a finitely generated projective A module over a local ring. The third section which forms the main part of the project, conducts a detailed study on projective modules of rank 1. The last section deals with stably free modules. There exist stably free modules which are not free. Therefore, these ‘stable’ objects are interesting to study.

1 Projective Modules

We shall refer A as a commutative ring with 1, unless it is specified.

Definition 1.1. An A - module P is called projective if there exists a module Q so that the direct sum $P \oplus Q$ is free.

Theorem 1.1. (Standard characterizations of a Projective Module)

Let P be an A - module. The following are equivalent:

1. For every diagram of A - modules

$$\begin{array}{ccc} & P & \\ & \downarrow f & \\ M & \xrightarrow{h} & N \longrightarrow 0 \end{array}$$

with exact line, there exist a homomorphism $g : P \rightarrow M$ called the lift of f making the diagram commutative;

2. for every epimorphism $f : M \rightarrow P$ there exists a $s : P \rightarrow M$ such that $f \circ s = id_P$;
3. P is a direct summand of a free A - module.

Proof. (1) $\implies$ (2). Take $N = P$ and $f = id_P$.

(2) $\implies$ (3). Every A - module P is a quotient of a free A - module F . Therefore there exist a surjective homomorphism f from F to P . Then by (2), there exist an $s : P \rightarrow F$ with $f \circ s = id_P$. Let $y \in F$, then $y = s(x) + y - s(x)$ where $x = f(y)$. Also $y - s(x) \in \ker f$ since $f(y - s(x)) = f(y) - f(s(x)) = x - x = 0$. Therefore $F = s(P) + \ker f$. Now, if $a \in s(P) \cap \ker f$, then $a = s(p)$ for some $p \in P$, therefore $0 = f(a) = f(s(p)) = p$ implies $a = 0$. Therefore

$$F = s(P) \oplus \ker f.$$

Since $s(P) \cong P$, $F \cong P \oplus \ker f$. Therefore P is a direct summand of a free module.

(3) $\implies$ (1). Let $P \oplus Q = F$, where F is a free A - module with basis B . We can define a map $f^* : F = P \oplus Q \rightarrow N$, $(p, q) \mapsto f(p)$. Since f is surjective the restriction $f^*|_B$ can be lifted to a map $\gamma : B \rightarrow M$ (for any $f^*(b) = n$, $b \in B$ & $n \in N$, $\exists m \in M$ such that $h(m) = n$, define $\gamma(b) = m$.) The linear extension $g' : F \rightarrow M$ is a lift of f^* . Then $g = g'|_P$ is the required lift of f . ■

Theorem 1.2. For a projective A -module P the following properties are equivalent:

1. P is finitely generated;
2. P is a direct summand of a finitely generated free module;
3. for some $n \in \mathbb{N}$, there is an exact sequence

$$A^n \rightarrow A^n \rightarrow P \rightarrow 0.$$

Proof. (1) $\implies$ (2). Since P is finitely generated, there exist an epimorphism $f : A^n \rightarrow P$. By the definition of projective module we have an $s : P \rightarrow A^n$ such that $f \circ s = id_P$. Let $y \in A^n$, then $y = s(x) + y - s(x)$ where $x = f(y)$. Also $y - s(x) \in \ker f$ since $f(y - s(x)) = f(y) - f(s(x)) = x - x = 0$. Therefore $A^n = s(P) + \ker f$. Now, if $a \in s(P) \cap \ker f$, then $a = s(p)$ for some $p \in P$, therefore $0 = f(a) = f(s(p)) = p$ implies $a = 0$. Therefore

$$A^n = s(P) \oplus \ker f.$$

Since $s(P) \cong P$, $A^n \cong P \oplus \ker f$. Therefore P is a direct summand of a finitely generated free module.

(2) $\implies$ (3). Let $A^n = P \oplus Q$. Let $\pi : A^n \rightarrow Q$ and $i : Q \rightarrow A^n$ be the natural projection and canonical embedding respectively. Then $i \circ \pi$ is a homomorphism from A^n to A^n whose image Q is the kernel of the projection $A^n \rightarrow P$. Thus we get the required exact sequence

$$A^n \xrightarrow{i \circ \pi} A^n \rightarrow P \rightarrow 0.$$

(3) $\implies$ (1). From the exact sequence we get a surjection $f : A^n \rightarrow P$. Therefore P is finitely generated. ■

Remarks:

1. A direct sum of two projective modules is projective.
2. If the ring A is commutative and M and N are projective A -modules, then so is $M \otimes_A N$. This is because if M and N are two projective A -modules, then $M \oplus P_1 \cong F_1$ and $N \oplus P_2 \cong F_2$ for some free A -modules F_1 and F_2 . Then

$$\begin{aligned} F_1 \otimes_A F_2 &= (M \oplus P_1) \otimes_A (N \oplus P_2) \\ &\cong (M \otimes_A N) \oplus (P_1 \otimes_A N) \oplus (M \otimes_A P_2) \oplus (P_1 \otimes_A P_2). \end{aligned}$$

Therefore $M \otimes_A N$ is the direct summand of a free A -module $F_1 \otimes_A F_2$.

3. If M is a finitely generated projective A -module, then $\text{Hom}_A(M, A)$ is also a finitely generated projective A -module. For: If M is a finitely generated

projective A - module, then $M \oplus N \cong A^n$ for some n . Now,

$$\begin{aligned} \operatorname{Hom}_A(M, A) \oplus \operatorname{Hom}_A(N, A) &\cong \operatorname{Hom}_A(M \oplus N, A) \\ &\cong \operatorname{Hom}_A(A^n, A) \\ &\cong A^n. \end{aligned}$$

We can associate the isomorphism classes of finitely generated projective module a monoid and semiring structures. Let A be a commutative ring. The isomorphism classes of finitely generated projective A -modules form a commutative monoid (semi-group with a identity element 0), the addition being defined by

$$[P] + [Q] = [P \oplus Q].$$

If we define the multiplication to be

$$[P] \cdot [Q] = [P \otimes_A Q]$$

then the isomorphism classes have a semiring structure.

2 Projective Modules over Local Rings

Here we prove an important result concerning the freeness of finitely generated projective modules over local rings. For that, we need some elementary results from commutative algebra which we recall below.

Definition 2.1. The **Jacobson radical** $Jac(A)$ of a ring A is the intersection of all the maximal ideals of A .

Lemma 2.1. An element $x \in A$ belongs to $Jac(A)$ if and only if $1 - ax$ is a unit for all $a \in A$.

Proof. Let $x \notin Jac(A)$. Then for every $a \in A$, ax belongs to every maximal ideal, so that $1 - ax$ does not belong to any maximal ideal. Therefore the ideal $A(1 - ax)$ generated by $1 - ax$ is A . Therefore $(1 - ax)$ is a unit.

Conversely if $x \notin Jac(A)$, there is a maximal ideal $\mathfrak{m}$ with $x \notin \mathfrak{m}$. Then $\mathfrak{m}$ and x generate the unit ideal (1) , so that we have $u + xy = 1$ for some $u \in \mathfrak{m}$ and some $y \in A$. Hence $1 - xy \in \mathfrak{m}$ and is therefore not a unit. ■

Theorem 2.1. (Nakayama's lemma) Let $\mathfrak{a}$ be an ideal contained in Jacobson radical of ring A , and let M be an A -module. If M is a finitely generated A -module and $\mathfrak{a}M = M$ then $M = 0$.

Proof. Let $x_1, x_2, \dots, x_n$ generate M . Choose the least integer n with this property. Assume that $n \geq 1$. Since $x_n \in \mathfrak{a}M$, $x_n = a_1x_1 + \dots + a_nx_n$ where $a_i \in \mathfrak{a}$. Then we get, $(1 - a_n)x_n = a_1x_1 + \dots + a_{n-1}x_{n-1}$. Now since, a_n is in Jacobson ideal, $1 - a_n$ is a unit. This means that $x_n \in \text{span}\{x_1, x_2, \dots, x_{n-1}\}$ which contradicts the minimality of n . Therefore, $n = 0$. i.e, $M = 0$. ■

Corollary 2.1. *If $N + \mathfrak{a}M = M$, then $M = N$.*

Proof. We have $\mathfrak{a}(M/N) = (N + \mathfrak{a}M)/N = M/N$, Now by Nakayama's lemma, we have $M/N = 0$, i.e. $M = N$. ■

Theorem 2.2. *Let A be a ring, $\mathfrak{a}$ an ideal, M an A -module. Then $A/\mathfrak{a} \otimes_A M$ is isomorphic to $M/\mathfrak{a}M$.*

Proof. Tensoring the exact sequence

$$0 \rightarrow \mathfrak{a} \rightarrow A \rightarrow A/\mathfrak{a} \rightarrow 0$$

with M yields an exact sequence

$$\mathfrak{a} \otimes_A M \xrightarrow{j} A \otimes_A M \xrightarrow{i} A/\mathfrak{a} \otimes_A M \rightarrow 0.$$

By the surjectivity of i ,

$$A/\mathfrak{a} \otimes_A M \cong (A \otimes_A M)/\ker i \cong (A \otimes_A M)/\text{im } j \cong M/\text{im } j.$$

But $\text{im } j = \mathfrak{a}M$ implies $A/\mathfrak{a} \otimes_A M \cong M/\mathfrak{a}M$. ■

Theorem 2.3. *Every finitely generated projective A - module over a local ring A is free.*

Proof. Let P be a finitely generated projective A - module. Let $\mathfrak{m}$ be the maximal ideal of A and $k = A/\mathfrak{m}$ be the residue field. Let $a_1, \dots, a_n$ be the elements of P whose residue classes form a basis for the k - vector space $P/\mathfrak{m}P$. Then P is generated by $\{a_1, \dots, a_n\} \cup \mathfrak{m}P$. Let N be the submodule of P generated by $\{a_1, \dots, a_n\}$. Then $N + \mathfrak{m}P = P$. Therefore by Nakayama's lemma, we have $N = P$. i.e. P is generated by $\{a_1, \dots, a_n\}$. Therefore we have the exact sequence

$$0 \rightarrow Q \rightarrow A^n \xrightarrow{f} P \rightarrow 0$$

where f is defined by $e_i \mapsto a_i$ and Q is the kernel of f . Since P is projective this sequence splits and Q is finitely generated. We have $A^n \cong P \oplus Q$. Also this sequence

remain exact after tensoring with Q . i.e.

$$0 \rightarrow k \otimes_A Q \rightarrow k \otimes_A A^n \xrightarrow{k \otimes_A f} k \otimes_A P \rightarrow 0$$

is exact. But $k \otimes_A A^n \cong k^n$ and by construction $k \otimes_A f$ is an isomorphism. Therefore $k \otimes_A Q = 0$. By 2.2, $k \otimes_A Q = Q/\mathfrak{m}Q$. Therefore $Q/\mathfrak{m}Q = 0$ or $Q = \mathfrak{m}Q$. Again by Nakayama's lemma $Q = 0$. Therefore $P \cong A^n$. ■

Definition 2.2. An A -module M is called finitely presented, if and only if there are $m, n \in \mathbb{N}$ and an exact sequence of module homomorphisms

$$A^m \xrightarrow{\alpha} A^n \rightarrow M \rightarrow 0.$$

Informally, M is a finitely presented A -module if and only if it is finitely generated and the module of relations among these generators is finitely generated as well.

Theorem 2.4. *If M is a finitely generated A -module and $\mathfrak{p} \in \text{spec}(A)$ such that $M_{\mathfrak{p}} = 0$, then there exists an $s \in A - \mathfrak{p}$ such that $M_s = 0$.*

Proof. Let M be generated by the elements $x_1, \dots, x_n$. Since $M_{\mathfrak{p}} = 0$, there exist $s_1, \dots, s_n$ with $s_i x_i = 0$. Then choose $s = s_1 \dots s_n \notin \mathfrak{p}$, we get $M_s = 0$. ■

Theorem 2.5. *Let A be a commutative ring and P be a finitely presented A -module. The following properties are equivalent.*

1. P is projective;
2. For every maximal ideal $\mathfrak{m}$ of A , the $A_{\mathfrak{m}}$ module $P_{\mathfrak{m}}$ is free;
3. There are $f_1, \dots, f_n \in A$ such that $A = \sum A f_i$ and P_{f_i} is free over A_{f_i} for every i .

Proof. (1) $\implies$ (2). The ring $A_{\mathfrak{m}}$ is a local ring. By theorem 2.3, every finitely generated projective module over a local ring is free. Therefore $P_{\mathfrak{m}}$ is free.

(2) $\implies$ (3). Let $\mathfrak{m}$ be a maximal ideal of A . Then $P_{\mathfrak{m}}$ has a finite basis of the form $x_1/1, \dots, x_n/1$. Let $A^n \rightarrow P$ be defined by $x_1, \dots, x_n$ and C its cokernel. Since $P_{\mathfrak{m}} \cong A_{\mathfrak{m}}^n$, we have $C_{\mathfrak{m}} = 0$. Therefore by theorem 2.4, we have an $f \in A - \mathfrak{m}$ with $C_f = 0$. So we have an epimorphism $A_f^n \rightarrow P_f$. For its kernel K , we have $K_{\mathfrak{m}} = 0$ and K is finitely generated. Thus we get a $g \in A - \mathfrak{m}$ with $K_g = 0$. So P_{fg} is free ($C_{fg} = 0$ and $K_{fg} = 0$). In this way we get maybe an infinite family $(f_i)_{i \in I}$ such that P_{f_i} is free over A and to every maximal ideal $\mathfrak{m}$ there is an i with $f_i \notin \mathfrak{m}$. Therefore $\sum A f_i$ must be A since otherwise the ideal generated by all these elements will be contained in a maximal ideal which is not possible. Therefore since $1 \in A$, we get a finite subset of I such that $\sum_{i=1}^n A f_i = A$ and P_{f_i} is free over A_{f_i} .

(3) $\implies$ (2). Let $\mathfrak{m}$ be a maximal ideal of A . Since $\sum Af_i = A$, there exist an f_i such that $f_i \notin \mathfrak{m}$. So $P_{\mathfrak{m}}$ as a localization of P_{f_i} is free over $A_{\mathfrak{m}}$.

(2) $\implies$ (1). Let $M \rightarrow M'$ be an epimorphism. To show that P is projective, it is enough to show that the induced homomorphism $\text{Hom}_A(P, M) \rightarrow \text{Hom}_A(P, M')$ is an epimorphism. This is equivalent to saying that $(\text{Hom}_A(P, M))_{\mathfrak{m}} \rightarrow (\text{Hom}_A(P, M'))_{\mathfrak{m}}$ is an epimorphism for all maximal ideals $\mathfrak{m}$ of A . But since P is finitely presented, $(\text{Hom}_A(P, M))_{\mathfrak{m}} \cong \text{Hom}_A(P_{\mathfrak{m}}, M_{\mathfrak{m}})$. By the projectiveness of $P_{\mathfrak{m}}$ we get the epimorphism $\text{Hom}_A(P_{\mathfrak{m}}, M_{\mathfrak{m}}) \rightarrow \text{Hom}_A(P_{\mathfrak{m}}, M'_{\mathfrak{m}})$. Thus the induced homomorphism above is an epimorphism. ■

Lemma 2.2. Let $A \not\cong 0$ be a commutative ring. Then $A^m \cong A^n$ implies $m = n$.

Proof. Let $\mathfrak{m}$ be a maximal ideal of A , $k = A/\mathfrak{m}$ and let $\phi : A^m \rightarrow A^n$ be an isomorphism. Then $1 \otimes_A \phi : (A/\mathfrak{m}) \otimes_A A^m \rightarrow (A/\mathfrak{m}) \otimes_A A^n$ is an isomorphism. This implies $k^m \cong k^n$ since $k \otimes_A A^m \cong k^m$. Considering k^m and k^n as k -vector spaces we have $m = n$. ■

3 Rank of a Projective Module

Definition 3.1. Let $A \not\cong 0$ be a ring. The rank of a finitely generated free module A^m is defined as $rk_A(A^m) = rk(A^m) = m$. Let P be a finitely generated projective A -module. Then we define the **rank** map $rk(P) : \text{Spec}(A) \rightarrow \mathbb{N}$ by $rk(P)(\mathfrak{p}) = rk_{A_{\mathfrak{p}}}(P_{\mathfrak{p}})$.

Theorem 3.1. Let P be a finitely generated projective A -module. The map $rk(P) : \text{Spec}(A) \rightarrow \mathbb{N}$ is locally constant.

Proof. Let $\mathfrak{p} \in \text{Spec}(A)$. By theorem 2.5, there exist an $f \in A - \mathfrak{p}$ such that P_f is free over A_f . Let

$$D(f) = \{\mathfrak{p} \in \text{Spec}(A) : f \notin \mathfrak{p}\}$$

be the principal open subset associated with f . For every $\mathfrak{q} \in D(f)$, considering $P_{\mathfrak{p}}$ and $P_{\mathfrak{q}}$ as a localization of P_f we get $rk_{\mathfrak{p}}(P) = rk_{\mathfrak{q}}(P)$. Therefore the map is locally constant. ■

Theorem 3.2. Let I be an ideal of a commutative ring A contained in the Jacobson radical of A . If P and P' are two finitely generated projective A -modules such that $P/IP \cong P'/IP'$. Then $P \cong P'$.

Proof. Since $P/IP \cong P'/IP'$, we have two epimorphisms $P \rightarrow P'/IP'$ and $P \rightarrow P'/IP'$. By the projectivity of P we have the commutative diagram

$$\begin{array}{ccc}
& & P \\
& \swarrow f & \downarrow \pi \\
P' & \xrightarrow{h} & P'/IP'
\end{array}$$

such that $h \circ f = \pi$. Then $f(P) + IP' = P$. Because, there exist an element $p \in P$ for any element $p' \in P'$, such that $h(p') = \pi(p) = h(f(p))$ by the surjectivity of π and the commutativity of the diagram. Therefore $p' - f(p) \in \ker h = IP'$, which implies there exist an element $i \in I$ such that $p' = f(p) + ip'$. Thus $P' \subset f(P) + IP'$ and the other inclusion is trivial. Now by Nakayama's lemma we have $f(P) = P$. i.e. f is surjective.

By the projectivity of P' , we see $\ker f$ is finitely generated and is a direct summand of P . It is clear from the commutative diagram

$$\begin{array}{ccc}
P & \xrightarrow{f} & P' \\
\downarrow \Pi_1 & & \downarrow \Pi_2 \\
P/IP & \xrightarrow{\bar{f}} & P'/IP'
\end{array}$$

the kernel $\ker f / I \ker f$ of $\bar{f}$ is zero since $\bar{f}$ is an isomorphism. Therefore by Nakayama's lemma, we have $\ker f = 0$. Thus $P \cong P'$. $\blacksquare$

4 Projective Modules of Rank 1

Here we present some equivalent conditions for a projective A -module to be of rank 1.

Theorem 4.1. *Let A be a commutative ring and M be a finitely presented A -module. The following statements are equivalent.*

1. *The evaluation map $M \otimes_A M^* \rightarrow A$, $(x, \alpha) \mapsto \alpha(x)$ is an isomorphism;*
2. *There exists an A -module N such that $M \otimes_A N \cong A$;*
3. *$M_{\mathfrak{m}} \cong A_{\mathfrak{m}}$ as $A_{\mathfrak{m}}$ modules for every maximal ideal $\mathfrak{m}$ of A ;*
4. *$M_{\mathfrak{p}} \cong A_{\mathfrak{p}}$ for every $\mathfrak{p} \in \text{Spec}(A)$;*
5. *M is projective of rank 1.*

Proof. (1) $\implies$ (2). Take $N = M^*$.

(2) $\implies$ (3). For every maximal ideal $\mathfrak{m}$ we have $M_{\mathfrak{m}} \otimes_{A_{\mathfrak{m}}} N_{\mathfrak{m}} \cong A_{\mathfrak{m}}$. Since $A_{\mathfrak{m}}$ is local, without loss of generality we can assume A is a local ring with maximal ideal $\mathfrak{m}$ and residue field $k = A/\mathfrak{m}$. We have $M \otimes_A N \cong A$ hence $(M \otimes_A N) \otimes_A A/\mathfrak{m} \cong A/\mathfrak{m}$. i.e. $M \otimes_A (N \otimes_A A/\mathfrak{m}) \cong M \otimes_A N/\mathfrak{m}N$ and consequently $M/\mathfrak{m}M \otimes_A N/\mathfrak{m}N \cong A/\mathfrak{m}$.

As a result, we get this as an isomorphism of k - vector spaces. Since $A/\mathfrak{m} = k$ is a one dimensional vector space, the vector space $M/\mathfrak{m}M \cong k$. Therefore $M/\mathfrak{m}$ is generated by a single element say the residue class of $x \in M$. Therefore x and $\mathfrak{m}$ generate M and by Nakayama's lemma, we have M is generated by a single element. So there is an epimorphism $A \rightarrow M$ which implies $M \cong A/I$ for some ideal I of A . Thus $M \otimes_A N \cong A/I \otimes_A N \cong N/IN$. Therefore $I(M \otimes_A N) = 0$. But $M \otimes_A N \cong A$ yields $I = 0$. Thus we get $M_{\mathfrak{m}} \cong A_{\mathfrak{m}}$.

(3) $\implies$ (4). Let $\mathfrak{p} \in \text{Spec}(A)$. Then $\mathfrak{p} \subset \mathfrak{m}$ for some maximal ideal $\mathfrak{m}$. Then $M_{\mathfrak{p}} = [M_{\mathfrak{m}}]_{\mathfrak{p}} \cong [A_{\mathfrak{m}}]_{\mathfrak{p}} = A_{\mathfrak{p}}$.

(4) $\implies$ (3). This follows easily since every maximal ideal is a prime ideal.

(3) $\implies$ (5). Since $M_{\mathfrak{m}}$ is $A_{\mathfrak{m}}$ free, by theorem 2.5, we have M is projective. Since $M_{\mathfrak{p}} \cong A_{\mathfrak{p}}$ for every $\mathfrak{p} \in \text{Spec}(A)$ we have M is of rank 1.

(5) $\implies$ (1). We have $M_{\mathfrak{m}} \cong A_{\mathfrak{m}}$ for every $\mathfrak{m}$, $M_{\mathfrak{m}} \otimes_{A_{\mathfrak{m}}} M_{\mathfrak{m}}^* \cong A_{\mathfrak{m}} \otimes_{A_{\mathfrak{m}}} M_{\mathfrak{m}}^* \cong M_{\mathfrak{m}}^*$. But $M_{\mathfrak{m}} \cong A_{\mathfrak{m}}$ implies $M_{\mathfrak{m}}^* \cong A_{\mathfrak{m}}^*$ and $A_{\mathfrak{m}}^* \cong A_{\mathfrak{m}}$. Therefore we get, $M_{\mathfrak{m}} \otimes_{A_{\mathfrak{m}}} M_{\mathfrak{m}}^* \cong A_{\mathfrak{m}}$ for every maximal ideal $\mathfrak{m}$. Thus $M \otimes_A M^* \cong A$. $\blacksquare$

As a consequence of this theorem we can associate the isomorphism classes of projective A -modules of rank 1 an abelian group structure by defining the operation as $[P].[Q] = [P \otimes_A Q]$ for the projective A - modules P and Q . The identity element is $[A]$ and the inverse of $[P]$ is $[P^*]$.

Definition 4.1. The group of isomorphism classes of projective A - modules of rank 1 is called the **Picard group** of A and is denoted by $\text{Pic } A$.

Definition 4.2. Let $Q(A)$ denote the ring of fractions of A , i.e. $Q(A) = S^{-1}A$ where S is the set of non-zero divisors of A . A **fractional ideal** of A is an A -submodule of $Q(A)$ for which there exists an $s \in S$ such that $sA \subset I \subset t^{-1}A$.

Definition 4.3. Let I, J be fractional ideals of A . Then define:

$$I : J = \{x \in S^{-1}A : xJ \subset I\} \quad \text{and} \quad I^{-1} = A : I.$$

It is easy to observe that if I, J are fractional ideals of A , then so are $I : J$ and IJ .

Theorem 4.2. Let I, J be fractional ideals of A . The map

$$\phi : I : J \rightarrow \text{Hom}_A(J, I)$$

defined by $x \mapsto (j \mapsto xj)$ is an isomorphism of A - modules.

Proof. Clearly ϕ is an A -module homomorphism. Now suppose $\phi(x) = 0$, then $xj = 0$ for every $j \in J$. Since J contains a non-zero divisor, $x = 0$. Therefore ϕ is injective.

Now let $\alpha; J \rightarrow I \in \text{Hom}_A(J, I)$ and $s \in S$ with $sA \subset J \subset s^{-1}A$. Set $x = s^{-1}\alpha(s)$. Now for any $j \in J$

$$\alpha(j) = s^{-2}(s^2\alpha(j)) = s^{-2}\alpha(s^2j) = s^{-2}\alpha(sjs) = s^{-2}sj\alpha(s) = xj$$

since $sj \in A$. Therefore ϕ is surjective. ■

Remark: $II^{-1} \subset A$.

Definition 4.4. An **invertible ideal** of ring A is a fractional ideal I such that $II^{-1} = A$.

Theorem 4.3. Let I be a fractional ideal of a commutative ring A . The following properties of I are equivalent:

1. I is invertible;
2. I is projective as an A -module;
3. I is projective of rank 1.

Proof. (1) $\implies$ (2). Since $1 \in II^{-1}$, there exist $a_1, \dots, a_n \in I$ and $b_1, \dots, b_n \in I^{-1}$ such that $\sum_{i=1}^n a_i b_i = 1$. Define

$$f : A^n \rightarrow I \quad e_i \mapsto a_i \quad \text{and}$$

$$g : I \rightarrow A^n \quad x \mapsto (b_1 x, \dots, b_n x).$$

Then for $x \in I$, $fg(x) = \sum a_i b_i x = x = id_I$. Therefore f is surjective and f splits. Thus I is a direct summand of A^n , i.e. I is projective.

(2) $\implies$ (1). I is isomorphic to some $A^{(E)}$ for some indexing set E . Therefore there exist homomorphisms $f : A^{(E)} \rightarrow I \quad e_i \mapsto a_i$ and $g : I \rightarrow A^{(E)} \quad x \mapsto \sum g_i(x) e_i$ such that $fg = id_I$. Now by theorem 4.2, we have for every $i \in E$, there exist $b_i \in I^{-1} \cong \text{Hom}_A(I, A)$ such that $g_i(x) = b_i x$ for every $x \in I$. From the definition of g it is clear that for every $x \in I$, $g_i(x) = b_i x = 0$ for all but finitely many i . Since there exist a unit of $S^{-1}A$ in I , $b_i = 0$ for all but finitely many i . Now

$$x = fg(x) = \sum a_i b_i x$$

for $x \in I$. Since some of these x are units, we have $1 = \sum a_i b_i$ with $a_i \in I, b_i \in I^{-1}$. Therefore $1 \in II^{-1}$ implies I is invertible.

(3) $\implies$ (1). It follows from (2) $\implies$ (1).

(1) $\implies$ (3). Since I is invertible, $II^{-1} = A$. If J is an ideal of A and E be a flat (in particular projective) A -module, then the map $\phi : J \otimes E \rightarrow JE$, $j \otimes e \mapsto je$ is an isomorphism. Therefore $I \otimes I^{-1} \cong II^{-1} = A$. By theorem 4.1, we have I is projective of rank 1. $\blacksquare$

Remark Let I, J be fractional ideals of A . The product IJ is the additive group generated by all products ab with $a \in I, b \in J$.

Definition 4.5. Let I, J be fractional ideals of A . Then $\mathbf{Inv}(A)$ denote the group of invertible ideals of A with respect to the above product IJ .

Let T be a multiplicatively closed subset of A containing non-zero divisors of A . Let $\mathbf{Inv}(A, T)$ denote the subgroup of $\mathbf{Inv}(A)$ consisting of such invertible ideals I of A , that $tA \subset I \subset t^{-1}A$ for some $t \in T$.

Definition 4.6. Let $\mathbf{Prin}(A)$ denote the group of invertible principal ideals of A , i.e. which are generated as A -modules by a single unit of $Q(A)$.

$\mathbf{Prin}(A, T) = \mathbf{Inv}(A, T) \cap \mathbf{Prin}(A)$.

Theorem 4.4. With the above notations, there is a canonical exact sequence

$$0 \rightarrow \mathbf{Prin}(A, T) \xrightarrow{\alpha} \mathbf{Inv}(A, T) \xrightarrow{\beta} \mathbf{Pic}(A) \xrightarrow{\pi} \mathbf{Pic}(T^{-1}A).$$

Proof. For $I \in \mathbf{Inv}(A, T)$, $\beta(I) = [I]$. Then $\ker \beta = \{I \in \mathbf{Inv}(A, T) \mid [I] = [A]\} = \{I \in \mathbf{Inv}(A, T) \text{ and } I \in \mathbf{Prin}(A)\} = \mathbf{Prin}(A, T)$.

Therefore now only the exactness at $\mathbf{Pic}(A)$ is non-trivial. So we need to prove that $\text{im } \beta = \ker \pi$. Let $I \in \mathbf{Inv}(A, T)$. Then there is a $t \in T$ such that $tA \subset I \subset t^{-1}A$ so that

$$T^{-1}A = T^{-1}tA \subset T^{-1}I \subset T^{-1}t^{-1}A = T^{-1}A.$$

i.e. $T^{-1}A = T^{-1}I$. So $\beta(I) = [I] \in \ker \pi$. Therefore $\text{im } \beta \subseteq \ker \pi$.

Let $[P] \in \mathbf{Pic}(A)$ belong to $\ker \pi$. i.e. $T^{-1}P \cong T^{-1}A$. Let g be that isomorphism. Then there exist $p \in P, t \in T$ such that $g(p/t) = 1$ in $T^{-1}A$. We have the monomorphism $f : P \rightarrow T^{-1}P$, $p \mapsto p/1$. But $T^{-1}P \cong T^{-1}A$, so that $g(f(P)) = I$ is a submodule of $T^{-1}A$. Since $g(p/t) = 1$, we have $g(p/1) = t/1$. i.e. $t/1 = g(f(p))$. so by the definition of I , $t \in I$. We have $P \cong I$ by $g \circ f$. Since $P \cong I$ is finitely generated, say generated by $a_1/t_1, \dots, a_n/t_n$, $a_i \in A, t_i \in T$, we can choose $s = t_1 t_2 \dots t_n$ so that $s \in T$ and $sA \subset I \subset s^{-1}A$. Thus $I \in \mathbf{Inv}(A, T)$ and $\beta(I) = [I] = [P] \in \ker \pi$. Therefore $\ker \pi \subseteq \text{im } \beta$. $\blacksquare$

Example 1. Here we give an example of a non-free projective module of rank 1. Let $A = \mathbb{Z}(\sqrt{-5}) = \{a + b(\sqrt{-5}) \mid a, b \in \mathbb{Z}\}$. The ring of fractions $Q(A)$ of A is $\mathbb{Q}(\sqrt{-5})$. Consider the ideal $\mathfrak{a} = \langle 3, 1 + \sqrt{-5} \rangle$ of A . We claim that $\mathfrak{a}$ is invertible.

For: An element $(a+b\sqrt{-5}) \in \mathbb{Q}(\sqrt{-5})$ belongs to $\mathfrak{a}$ iff $3(a+b\sqrt{-5}) = 3a+3b\sqrt{-5} \in A$ and $(1+\sqrt{-5})(a+b\sqrt{-5}) = a-5b+(a+b)\sqrt{-5} \in A$. i.e. iff $3a, 3b, a-5b, a+b \in \mathbb{Z}$. If we write $a = a'/3$, $b = b'/3$ $a', b' \in \mathbb{Z}$, then the above condition gives the relations $a' + b' \equiv 0 \pmod{3}$ and $a' - 5b' \equiv 0 \pmod{3}$. But if $a' + b' \equiv 0 \pmod{3}$, it automatically follows $a' - 5b' \equiv 0 \pmod{3}$. Therefore by a simple observation, we will get that $-1, (2+\sqrt{-5})/3 \in \mathfrak{a}^{-1}$ and $3(2+\sqrt{-5})/3 + (1+\sqrt{-5})(-1) = 2+\sqrt{-5} - (1+\sqrt{-5}) = 1 \in \mathfrak{a}\mathfrak{a}^{-1}$. Thus $\mathfrak{a}$ is invertible. Therefore by theorem 4.3, $\mathfrak{a}$ is a projective A - module of rank 1.

If $\mathfrak{a}$ was a free A - module it must be a principal ideal. But this is not the case.

For: Consider the norm $N : \mathbb{Z}(\sqrt{-5}) \rightarrow \mathbb{Z}$, $a + b\sqrt{-5} \mapsto a^2 + b^2$

Clearly N is multiplicative, i.e. $N(xy) = N(x)N(y)$. From the definition of N , it is clear that an element $x \in A$ is a unit iff $N(x) = \pm 1$. Therefore if $3 = xy$ for some $x, y \in A$ which are not units, then $N(xy) = N(3) = 9$ implies $N(x) = N(y) = 3$. But $N(x) = 3$ is not possible since there is no integer solution for $a^2 + 5b^2 = 3$. Therefore 3 is irreducible, and by a similar argument $1 + \sqrt{-5}$ is also irreducible. Therefore $\mathfrak{a}$ is not principal and thus not free.

5 Stably Free Modules

Definition 5.1. Let R be an arbitrary ring. An R -module P is called **stably free** (of type m) if $P \oplus R^m \cong R^n$ for some non-negative integers m and n .

Stably free modules form an important class of projective modules. Stably free modules need not be free. But all stably free modules are projective.

Remark: A projective module P is stably free of type m if and only if P is isomorphic to the kernel of a suitable epimorphism $f : R^n \rightarrow R^m$, which automatically splits.

For: If P is stably free of type m , then $P \oplus R^m \cong R^n$ for some m and n . Thus we have the exact sequence

$$0 \rightarrow P \rightarrow R^n \xrightarrow{f} R^m \rightarrow 0.$$

where f is the projection. By the projectivity of R^m , f automatically splits. Converse is trivial. Let us translate these into a matrix language. If M is the $m \times n$ matrix associated with f , then M is right invertible, i.e, there exists an $n \times m$ matrix N such that $MN = I_m$. (By I_m we denote the $m \times m$ identity matrix). Conversely any right invertible $m \times n$ -matrix M defines a finitely generated stably free (right R -module) P of type m , namely the solution space of M , $P = \{\alpha \in R^n \mid M\alpha = 0\}$. Thus the study of stably free modules can be thought as a study of right invertible R - matrices.

Example 2 (P.M. Cohn). Let V be a vector space over a field k with countably infinite dimension. Let R denote the ring $\text{End}_k(V)$. Since V is infinite dimensional, $V \oplus V \cong V$ as a k -module, but we may also regard V as an R -module. Then we have

$$R \cong \text{Hom}(V, V) \cong \text{Hom}(V \oplus V, V) \cong \text{Hom}(V, V) \oplus \text{Hom}(V, V) \cong R \oplus R.$$

where all the terms are considered as R -modules. Therefore $R \cong R \oplus R$.

The above example implies that $R^s \cong R^t$ need not imply $s = t$.

Definition 5.2. We say that a ring R satisfies the **invariant basis property** (IBP) if, for any integers $s, t \geq 0$, $R^s \cong R^t$ (as right modules) implies $s = t$.

Example 3. 1. A field satisfies IBP.

2. A commutative ring ($R \not\cong 0$) satisfies IBP. (lemma 2.2)

3. A Noetherian ring R satisfies IBP. Otherwise there is an isomorphism $g : R^m \rightarrow R^n$ with $m < n$. Let $\pi : R^n \rightarrow R^m$ be the projection. Then $f = g \circ \pi : R^n \rightarrow R^n$ is a surjective endomorphism. Since π has non-trivial kernel, f also has non-trivial kernel. Let $y \neq 0 \in \ker(f)$, by the surjectivity of f , $\exists x \in R^n$ such that $f(x) = y$ and consequently $x \in \ker(f^2)$ and $x \notin \ker(f)$. Therefore we get an infinite increasing sequence of ideals

$$\ker(f) \subsetneq \ker(f^2) \subsetneq \ker(f^3) \subsetneq \dots$$

which is not possible in Noetherian ring.

A stably free module need not be free.

Example 4. The R -module $T = \{(f, g, h) \in R^3 : xf + yg + zh = 0 \text{ in } R\}$, where R is the ring $\mathbb{R}[x, y, z]/(x^2 + y^2 + z^2 - 1)$ is stably free, but not free.

Proof. Define a homomorphism from $\phi : R^3 \rightarrow R$ as $(u, v, w) \mapsto (x, y, z) \cdot (u, v, w)$ the dot product of two vectors. Then,

$$\ker \phi = \{(f, g, h) \in R^3 \mid (f, g, h)(x, y, z) = 0\} = T.$$

We see ϕ is an epimorphism since $(x, y, z)(x, y, z) = x^2 + y^2 + z^2 = 1$ in R . Therefore we get $R^3 \cong T \oplus R$. Thus T is stably free.

But $T \not\cong R^2$. Suppose $T \cong R^2$, then T has an R -basis, say (l, m, n) and (a, b, c) .

Then $(x, y, z), (l, m, n), (a, b, c)$ form a basis for R^3 , so that the matrix

$$M = \begin{bmatrix} x & l & a \\ y & m & b \\ z & n & c \end{bmatrix}$$

is invertible and its determinant is a unit in R .

We can evaluate the elements of R at the points of a unit sphere

$$S^2 = \{(x_0, y_0, z_0) \in \mathbb{R}^3 \mid x_0^2 + y_0^2 + z_0^2 = 1\}.$$

A unit in R never vanishes on the sphere since if $a(x, y, z)b(x, y, z) = 1$ in R , then $a(x_0, y_0, z_0)b(x_0, y_0, z_0) = 1$ in $\mathbb{R}$. Therefore the map $S^2 \rightarrow R^3, v \mapsto (l(v), m(v), n(v))$ is a continuous vector field on S^2 (each of the components is continuous since l, m, n are polynomials) which is never vanishing since $\det(M)$ is a unit. But this is not possible; since by Hairy ball theorem in topology, every continuous vector field on a sphere vanishes at least once. Therefore $T \not\cong R^2$. ■

Theorem 5.1. *The kernel P of an epimorphism $f : R^n \rightarrow R^m$ is free iff f can be lifted to an isomorphism $f' : R^n \rightarrow R^m \oplus R^r$ for some r , such that $\pi \circ f' = f$, where π is the projection from $R^m \oplus R^r$ to R^m .*

Proof. Suppose the isomorphism f' exists. Then $P = \ker f \cong \ker \pi = R^r$. Conversely, suppose there exists an isomorphism $g : P \rightarrow R^r$ for some r . We may write $R^n = Q \oplus P$ such that the restriction of f to Q gives an isomorphism $f_0 : Q \rightarrow R^m$. Then $f_0 \oplus g : R^n \rightarrow R^m \oplus R^r$ is the desired isomorphism f' . ■

We can convert the above theorem into a matrix language. Let M denote $m \times n$ matrix associated with f and N be the $(m + r) \times n$ matrix associated with f' if f' exists. Let P be the projection matrix associated with π as well. Then $\pi f' = f$ implies that $PN = M$, i.e., M is a submatrix of N consisting of its first m rows. The condition that f' is an isomorphism implies that there exist a matrix N' of suitable order such that $NN' = I_{m+r}$ and $N'N = I_n$. So we can formulate the matrix theoretic analogue of the above theorem.

Theorem 5.2. *For any right invertible $m \times n$ matrix M , the (stably free) solution space of M is free iff M can be completed to an invertible matrix by adding a suitable number of rows.*

Definition 5.3. We say that $v = (v_1, \dots, v_n), v_i \in R$ is a unimodular row if

$$\sum_{i=1}^n Rv_i = R.$$

The set of all unimodular rows of length n is denoted by $Um_n(R)$.

A row $(v_1, \dots, v_n)$ is unimodular iff there exist $w_i \in R$ such that $\sum_{i=1}^n v_i w_i = 1$.

Theorem 5.3. *For any ring R , the following statements are equivalent.*

1. *Any finitely generated stably free R module is free;*
2. *Any finitely generated stably free R module of type 1 is free;*
3. *Any unimodular row over R can be completed to an invertible matrix by adding suitable number of new rows.*

Proof. (1) $\implies$ (2). It is obvious.

(2) $\implies$ (1). We proceed by induction on the type m of a stably free module. When $m = 1$ (1) is true by (2). Assume (1) holds for $m - 1$. Let P be m -stably free. Then $P \oplus R^m \cong R^n$. But $P \oplus R^m = (P \oplus R^{m-1}) \oplus R \cong R^n$. i.e, $P \oplus R^{m-1}$ is 1- stably free and hence it is free. Thus $P \oplus R^{m-1} \cong R^k$ for some k . By induction, we have P is free.

(2) $\implies$ (3). Let v be a unimodular vector over R . Then v can be considered as a $1 \times n$ right invertible matrix whose solution space is free by (2). Therefore this follows from the theorem 5.2.

(3) $\implies$ (2). Let P be a finitely generated stably free module of type 1, i.e, $P \oplus R \cong R^n$ for some n . Then consider the $1 \times n$ matrix corresponding to the projection $R^n \rightarrow R$. Since the projection is an epimorphism, the matrix, considering as a vector over R is unimodular which can be completed into an invertible matrix by (3). Hence (2) follows immediately from the theorem 5.2. ■

Definition 5.4. A ring R which satisfies the above conditions is called a **Hermite ring**.

Definition 5.5. A unimodular row over R which can be completed to an invertible matrix is called **completable**.

Theorem 5.4. *Let R be a commutative ring with 1. Every stably free R - module of rank 1 is isomorphic to R .*

Proof. Let P be a stably free R - module of rank 1. Then P is the solution space of a right invertible $n - 1 \times n$ matrix M . Let $b_1, \dots, b_n$ be the maximal minors of M , i.e, b_i is the determinant of the $(n - 1) \times (n - 1)$ submatrix obtained by the i th column of M . Claim: $b_1, \dots, b_n$ generate the unit ideal.

For: If not, there exists a maximal ideal $\mathfrak{m}$ of R which contains all the b_i . Consider the field $k = A/\mathfrak{m}$. Let $\bar{M}$ be the image of M in $M_{n-1 \times n}(k)$. Since M is right invertible $\bar{M}$ is also right invertible, so it has rank $n - 1$. Since the $(n - 1)$ rows of $\bar{M}$ is independent, we can complete it into an invertible matrix M' in $GL_n(k)$. But if we calculate the determinant of M' by expanding with respect to the last row, we

get $\det(M') = 0$ as all the maximal minors are in $\mathfrak{m}$. This is a contradiction.

Therefore there exists $a_1, \dots, a_n$ such that $\sum a_i b_i = 1$. Therefore adding $(a_1, \dots, a_n)$ with appropriate signs for a_i as the last row of M , we will have $\det(M) = 1$, i.e, $M \in GL_n(R)$. So we could complete M into an invertible matrix, which implies by theorem 5.2 that the solution space P of M is free. Since P is 1- stably free, we must have $P \cong R$. ■

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